

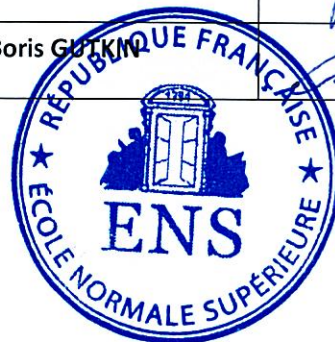
Nom et prénom du doctorant : BLANC Alexandre

Date de la soutenance : 5 décembre 2025

Président du Jury : Boris Gubkin

Rapport joint

Prénom et Nom	Signature	Prénom et Nom	Signature
Jean-Baptiste MASSON		Gordon BERMAN	
Adrienne FAIRHALL		Éric MOULINES	
Aleksandra WALCZAK		Boris GUBKIN	



Report Public Defence
Alexandre Blanc

Alexandre Blanc's public defence commenced with his providing a clear and engaging summary of his thesis, which is an extensive and thorough piece of work that analyzes the mechanisms of behavioral states in *Drosophila* larvae, from behavioral categorisation to neural mechanisms of neuro-modulation and its impact on behavior to mechanical modelling of the larvae body.

The thesis presentation started with a thorough and well laid out review of the background of his work, including information about *Drosophila* as a model organism in neuroscience, the methods and approaches of computational neuroscience, and the foundations of biomechanical modelling. This review showed the breadth and the depth of the candidate's knowledge of this multi-disciplinary field. The presentation proceeded to address the three topics of the thesis in the context of theory and modelling of behavior and the associated neuronal dynamics. First, he described the methods that he has contributed for characterizing behavioral variation, notably machine learning methods for annotation of behavior and the occupation statistics of behavioral states. This was followed by a presentation of his work on developing a firing rate model of neuromodulation and how it may impart context dependence to multiple behavioral repertoires of the animal. The presentation then outlined a major piece of work on designing and implementing a biomechanical model of the larva body. The defence presentation concluded with future perspectives on neural and biomechanical modelling.

The reporters and the rest of the jury then took turns to ask questions. These were designed to test the breadth of his knowledge of the theory and the methodology of the topics discussed and in the written thesis. The jury challenged the candidate both on the technical level as well as on the interpretation of research that formed the foundation of the work. Alexandre Blanc answered these challenges in a clear and well-founded manner. In his answers, Alexandre demonstrated a broad knowledge of the theory and methods applied in the studies. His answers were very well informed and solidly grounded in literature and in his own results as they were also pedagogical. The candidate demonstrated masterful knowledge in a wide variety of disciplines that all went into the research project.

The jury appreciated the clarity and the depth of the presentation, and in particular the ability of the candidate to present this complex topic in a didactic and pedagogical manner.

The jury then conferred in camera, and arrived at the unanimous view that Alexandre Blanc should be awarded the PhD. A discussion ensued about whether the thesis could be published verbatim; the jury agreed that the thesis is ready to be published in its present form.

